APEX RETAIL GROUP | INDEPENDENT HOMEWORK ASSIGNMENT

DOCUMENT: CLASS HOMEWORK (SESSION 4)

SCENARIO: VENDOR SUPPLY CHAIN LOG AUDIT & METADATA DISCOVERY

ASSIGNMENT OVERVIEW: You are assisting the Supply Chain Director at Apex Retail Group. The company receives shipments from various vendors logged into a central tracking database.
The director wants an initial data health audit of recent vendor shipment logs to understand data structure, inspect column data types, evaluate non-null entry integrity, and summarize key shipment metrics (shipment weight, freight cost).

RAW VENDOR SHIPMENT LOG DATASET:

```Python
import pandas as pd

shipment_data = {
    "shipment_id": ["SHP-801", "SHP-802", "SHP-803", "SHP-804", "SHP-805", "SHP-806", "SHP-807", "SHP-808", "SHP-809", "SHP-810"],
    "vendor_name": ["ApexLogistics", "GlobalFreight", "ApexLogistics", "FastShip", "GlobalFreight", "FastShip", "ApexLogistics", "GlobalFreight", "FastShip", "ApexLogistics"],
    "destination_dc": ["DC-Alpha", "DC-Beta", "DC-Alpha", "DC-Gamma", "DC-Beta", "DC-Gamma", "DC-Alpha", "DC-Beta", "DC-Gamma", "DC-Alpha"],
    "weight_lbs": [1250, 3400, 2100, 850, 4200, 1100, 3100, 2900, 950, 2750],
    "freight_cost": [450.00, 1120.00, 720.00, 310.00, 1450.00, 400.00, 1050.00, 980.00, 340.00, 910.00],
    "delivery_status": ["Delivered", "Delivered", "In Transit", "Delivered", "Delayed", "Delivered", "In Transit", "Delivered", "Delivered", "In Transit"]
}
```

PROBLEM 1: DATAFRAME LOADING & STRUCTURAL INSPECTION

In a Jupyter Notebook named `homework_session4_yourname.ipynb`:

1. Load `shipment_data` into a DataFrame named `df_shipments`.
2. Display the total number of rows and columns using `.shape`.
3. Display the first 5 records (`.head()`) and last 3 records (`.tail(3)`).

PROBLEM 2: METADATA & STATISTICAL AUDIT

1. Print the output of `df_shipments.info()` to verify data types and non-null values.

2. Generate numerical summary statistics using `df_shipments.describe()`.
3. Calculate:
    - Total freight cost across all shipments (`.sum()`).
    - Average shipment weight in lbs (`.mean()`).
    - Maximum freight cost logged (`.max()`).

PROBLEM 3: DATASET HEALTH BRIEFING

Write a 150-word formal briefing memo to the Supply Chain Director containing:

1. Dataset structural summary (row count, column count, column names).
2. Key quantitative metrics (total freight expenditure, average shipment weight).
3. A statement on data health confirming whether all columns contain expected data types and if any structural defects were observed.

SUBMISSION DEADLINE: Prior to Session 5.